V.F.S. 3.6 The Turning Interior

Orientation, Passage, and the Knot That No Reckoning Shows

Clean edition

What this volume establishes

3.5 gave one interior its geometry: a hyperbolic three-space with a normalized Ricci flow, a canonical form, and a surgery. It described a *state*. This volume describes what that state cannot see about itself.

1. **A vessel is one subject with many positions.** At every moment it wills one thing and does one thing: exactly one facet is live. The others are conditions it may occupy, each holding what it has accumulated. Edges are not bonds between co-present parts; they are *passages*.
2. **A facet's eight numbers split five and three.** Read Voluntas and Factum as directions — the corpus glosses Voluntas as *inner orientation* — and a facet carries (V, F, σ, λ). Five of these are invariants and are retained in rest; three are the *frame*, and are carried by the living moment, not stored anywhere.
3. **A passage is a vector.** Each edge carries a rotation vector $\omega = \theta n \in \mathbb{R}^3$ — direction the axis of turn, length the angle — together with a rate. Choosing where to go *is* choosing how one will be turned; the two are one datum.
4. **Rotation vectors do not add, and that is the whole content.** A circuit of one's own positions returns the frame turned by the composite, which is not the sum. That failure to close is the **knot**.
5. **The knot is invisible where it acts and detectable only in a life.** No local step sees it: the rule that selects the next passage reads only the carried frame and the passages available. Yet where one is repeatedly taken differs, and that difference shows in the record.
6. **And the whole layer describes a divided condition.** The corpus's own law couples will and act *symmetrically*, so their directions converge: the frame degenerates and the knot goes latent. These phenomena hold while a life's willing and doing diverge, and lose their grip as the two come together — integration does not untie the knot, but leaves it nothing to act on.
7. **And it cannot be reached from within.** Interior labour — cleansing, terminus, folding — acts on the retained five. The knot lives in the carried three. They are complementary halves of the same eight numbers.

I. The object

1 One subject, many positions

Definition 1 (Vessel). *A vessel is a finite connected graph $\Gamma = (V, E)$ whose vertices are* facets *— conditions the life may occupy — and whose edges are* passages *between them. At each moment exactly one facet is* live.

▷ *Plain.* The multiplicity here is of positions and routes, never of simultaneous selves. A vessel is one who wills and acts, and the graph is the map of where that willing and acting may stand.

2 The facet, and what it keeps

Definition 2 (Facet data)**.** *A facet carries $V, F \in \mathbb{R}^3$ — will and act as directions — together with resistance σ and Sophia λ. Synergy is $u = \sqrt{V\cdot F + \varepsilon}$, the corpus's own formula read as an inner product.*

Proposition 1 (The split $8 = 5+3$)**.** *The pair (V,F) decomposes losslessly into three invariants $(|V|, |F|, V\cdot F)$ and a frame $Q = (e_1, e_2, e_1\times e_2)$, $e_1 = V/|V|$, $e_2 \propto F - (F\cdot e_1)e_1$, with inverse*

$$V = |V|\, e_1, \qquad F = \tfrac{V\cdot F}{|V|}\, e_1 + \sqrt{|F|^2 - \left(\tfrac{V\cdot F}{|V|}\right)^2}\; e_2.$$

A facet therefore holds five numbers in rest — $|V|, |F|, V\cdot F, \sigma, \lambda$, equivalently σ, u, λ and two more — while the three frame parameters are carried by the live moment and stored nowhere.

Proposition 2 (The interior of 3.5 is the invariant part)**.** *The coordinates (σ, u, λ) that 3.5 geometrizes are three of the five retained invariants. The interior is thus a faithful three-dimensional geometry of a coarser object than the facet: it discards the frame, and it compresses $|V|, |F|, V\cdot F$ into the single u. In particular u does not distinguish will and act* weakening *from will and act* diverging*, and neither does the fold at $u = \Lambda_c$.*

3 Where the frame must live

Theorem 1 (Forced placement)**.** *The frame cannot lie in any space in which the ascent is a coordinate.*

(i) Not among (σ, u, λ). *A rotation by θ gives $\dot\lambda' = \sin\theta\, \dot u + \cos\theta\, \dot\lambda$, and synergy may fall while Sophia rises; monotonicity fails for every $\theta \neq 0$.*

(ii) Not in the spatial slice of 3.0. *There the three spatial directions are the living surface (x, y) and the ascent λ, a block structure whose frame freedom is $SO(2)$, abelian.*

(iii) In (V, F). *A simultaneous rotation preserves $|V|, |F|, V\cdot F$, hence u, hence the entire interior evolution — provided the (V,F) law is $SO(3)$-equivariant. The three residual parameters are a full $SO(3)$:* $6 - 3 = 3$.

▷ *Math.* The placement is not a convenience. Every space in which the ascent appears as a direction has that direction distinguished by monotonicity, and a distinguished direction reduces $SO(3)$ to $SO(2)$. Orientation must therefore sit above the geometry, in the space of willing and acting, where no direction is the ascent.

Theorem 2 (Three factors, three contents)**.** *The isometries of $\mathbb{H}^3$ decompose as $K\cdot A\cdot N$, and each factor carries a distinct relational content:*

factor	*what it is*	*what it carries*
$A \cong (\mathbb{R}, +)$	*translations along a geodesic*	*the* relative rate *— 3.5's own datum*
$K = SO(3)$	*rotations fixing a point*	*the* frame*, hence the knot*
N	*unipotent*	*not licensed here (see §3)*

3.5 uses A alone, which is why it is abelian: translations along one geodesic commute, and no ordering of reconciliations can be expressed. The knot lives in K, where rotations do not commute, and ordering is its entire content. That A carries state and K carries connection also settles what the terminus can touch: the reset acts in A, and $A \cap K = \{e\}$.

▷ *Math.* A candidate group must contain 3.5's additive relative rate as a one-parameter subgroup. Every one-parameter subgroup of $SO(3)$ alone is periodic, so $SO(3)$ by itself cannot carry that datum; the recovery of 3.5 is restriction to A.

Proposition 3 (The licensed structure)**.** *The transformations compatible with the corpus's own laws are $SO(3)$ on the frame together with dilation and translations of the interior: a six-dimensional group, of which the* connection *requires four — $SO(3) \times \mathbb{R}$, compact times abelian. It contains no unipotent elements: every rotation is semisimple. Compactness is what makes the reconciliation floor attained, and what makes the Frobenius energy bi-invariant, hence canonical.*

Proposition 4 (Degeneracy under alignment)**.** *The frame is determined only when V and F are linearly independent. Where will and act are parallel, or exactly opposed, the pair has rank one and only an $SO(2)$ remains. A vessel whose will and act coincide in every facet therefore has no orientation anywhere, and no knot can appear in it whatever its topology.*

4 The passage

Definition 3 (Edge)**.** *A passage carries $(\omega, r) \in \mathbb{R}^3 \times \mathbb{R}$: a rotation vector $\omega = \theta n$ — axis n, angle θ — and a relative rate. The structure group is $SO(3) \times \mathbb{R}$, compact times abelian, of dimension four. Reversal is $-\omega$.*

Proposition 5 (Subdivision is invisible)**.** *Splitting a passage into consecutive pieces changes nothing: the composite is all that is ever seen, and $b_1 = |E| - |V| + 1$ is unchanged. A passage is an atom of the theory — not because it is simple, but because its internal structure is unobservable in principle. Only* adding *or* cutting *a passage alters b_1.*

Proposition 6 (Dimensions)**.** *A facet has eight numbers; a passage four; the whole vessel $8|V| + 3b_1 - 3$. The knot alone occupies the character variety, of dimension 1 for $b_1 = 1$ and $3b_1 - 3$ for $b_1 \geq 2$.*

Proposition 7 (How few facets)**.** *The corpus fixes only "finite connected". The phenomena fix minima. With one passage per pair of facets, an irreducible circuit needs* ***three*** *facets and the invisible knot* ***four****; allowing several passages between the same two facets lowers both to two. Together with degeneracy under alignment the condition is joint: a knot requires a circuit* and *at least two facets in which will and act diverge.*

II. The knot

5 Holonomy, and why it exists

Theorem 3 (Rotation vectors do not add)**.** *For passages ω_1, ω_2 the composite rotation vector is not $\omega_1 + \omega_2$, and depends on the order. Witnessed:* $(0.6, 0.2, -0.3)$ *and* $(-0.1, 0.7, 0.4)$ *compose* *to* $(0.324, 0.961, -0.119)$ *against a sum* $(0.5, 0.9, 0.1)$ *— a discrepancy of* 0.288*; reversing the order gives* $(0.613, 0.752, 0.319)$*, a further* 0.565 *away. The discrepancy is the commutator, and it vanishes only in the limit of small turns: relative error* 27.8%, 8.2%, 2.7%, 0.8%, 0.3% *at scales* $1, 0.3, 0.1, 0.03, 0.01$.

Corollary 1 (The knot)**.** *Traversing a circuit of facets returns the carried frame rotated by the holonomy* $\mathrm{Hol}_c = g_{e_k} \cdots g_{e_1}$*, while every retained invariant is unchanged. If* $\mathrm{Hol}_c \neq 1$*, the circuit does not close: this is the knot. It is a property of* routes*, not of any facet and not of the frame — the frame is what it acts on.*

▷ *Plain.* Gentle passages nearly add, and a life whose transitions are soft is nearly free of knots. Sharp passages compose badly, and there the knot has its strength. Nonlinearity here is a measure of abruptness.

6 Invisibility

Theorem 4 (The invisible knot)**.** *If the holonomy image is a perfect group, every abelian invariant of the connection vanishes: the induced abelian datum is trivial and its harmonic residual is zero, so the criterion of 3.5 reports* CANONICAL*. Yet the representation is nontrivial and gauge-invariantly so, and no interior operation trivializes it. The obstruction of 3.5 is thus necessary but not sufficient, strictly.*

Proposition 8 (Uniqueness of the type)**.** *Among the finite subgroups of* $SO(3)$ *— cyclic, dihedral,* A_4*,* S_4*,* A_5 *— exactly one is perfect: the icosahedral* A_5*, whose abelianization is trivial where the others give* $\mathbb{Z}/n$*,* $(\mathbb{Z}/2)^2$*,* $\mathbb{Z}/3$*,* $\mathbb{Z}/2$*. There is therefore exactly one* type *of finite invisible knot, and it is intrinsic to the vessel's own geometry.*

Proposition 9 (Partial invisibility is a ladder)**.** *The abelian reading detects* $G/[G,G]$ *and misses* $[G,G]$*. Among the binary polyhedral groups the missed fraction runs* 0*,* $1/4$*,* $1/3$*,* $1/2$*,* 1 *for cyclic,* Q_8*,* $2T$*,* $2O$*,* $2I$*. The invisible knot is the top of this ladder. There is no ladder of nilpotent* depth*: a group with trivial abelianization is perfect, hence its lower central series is stationary and no intermediate depth exists.*

7 How many

Proposition 10 (Counting)**.** *Knots in general form a continuum of dimension* 1 *or* $3b_1 - 3$*. Invisible knots are finite in number: at* $b_1 = 2$*, of* 3600 *pairs in* A_5*,* 2280 *generate, giving* 38 *classes up to conjugation and* 19 *up to* $\mathrm{Aut}(A_5)$*.*

Theorem 5 (Three families)**.** *Under Nielsen moves — the elementary re-cuttings of the cycle basis — the* 38 *classes form* ***three*** *components, of sizes* 18*,* 10*,* 10*. The outer automorphism fixes the large one and exchanges the two small ones. Hence there are three genuinely distinct invisible knots, two of them mirror images of each other, and no re-description identifies them.*

8 Sectors

Theorem 6 (Trace criterion)**.** *Since* $\|M \mp I\|_F^2 = \|M\|^2 \mp 2\,\mathrm{Re}\,\mathrm{tr}\,M + 2$*, flipping the sign of a generator leaves the reconciliation energy unchanged* as a function *precisely when* $\mathrm{Re}\,\mathrm{tr}(A_k) = 0$*.*

The number of incompatible reckonings is therefore 2^m*, where* m *counts the generators of non-vanishing real trace. A returning relation that is its own undoing — a half-turn — admits one reckoning; every other admits two.*

Theorem 7 (The traps belong to the group, not the web)**.** *Read one and the same connection on one and the same ring twice: with abelian phases, and with full rotations. The abelian reading has several local minima of the reconciliation energy; the nonabelian reading has exactly one. Witnessed on* C_6, C_8, C_{10}, C_{12}*: three, two, three, three minima over* $U(1)$ *against one over* $SU(2)$ *in every case. The mechanism is topological: over* $U(1)$ *the configurations split into sectors labelled by an integer winding, since* $\pi_1(U(1)) = \mathbb{Z}$*; over the simply connected* $SU(2)$ *there is no winding to be caught by, and the sectors merge.* ***Roughness of the landscape is carried by*** π_1 ***of the group of relational acts, not by the number of loops.***

Corollary 2 (Multiplicity is an artefact; obstruction is not)**.** *Enlarging the group collapses the* multiplicity *of minima but does not remove the* obstruction*: a positive floor stays positive. Witnessed: rings carrying a prescribed fractional holonomy give three and two minima over* $U(1)$ *with floors* 4.231 *and* 0.979*, and one minimum over* $SU(2)$ *with floors* 0.154 *and* 0.492*. A frustration that is a genuine obstruction is the abelian shadow of the knot and survives into this layer unchanged; only the several dead ends dissolve.*

Proposition 11 (Two-fold trapping)**.** $\pi_1(SO(3)) = \mathbb{Z}/2$*, and on a single loop the reconciliation energy has exactly two local minima, both genuinely reachable. Which reckoning a life labours within is fixed by where the labour began.*

Theorem 8 (No false canonization)**.** $E(h) = 0$ *if and only if the connection is genuinely trivializable. The structure group being compact, the infimum for a knotted connection is attained and strictly positive. A vessel may rest short of canonical form; no resting place ever registers as completion.*

III. The living passage

9 The dispatcher

Definition 4 (Selection)**.** *At a facet, the outgoing passage is selected by comparing the carried frame with the passages' own axes — for instance, the passage whose axis lies closest to* e_1*. The rule is equivariant: a global turn of everything changes nothing.*

Proposition 12 (Reconstitution on arrival)**.** *When the life reaches a facet, its will and act are rebuilt from that facet's* stored *invariants together with the* arriving *frame, by the inverse of the* 5+3 *split. Every stored invariant is reproduced exactly; only the facing is new. Witnessed across a three-cycle: stored* $|V| = 0.9000, 1.5000, 1.3000$ *and* $V \cdot F = 0.4000, 0.6000, 0.9800$ *all recovered to four decimals, with the frame returning rotated by the holonomy.*

▷ *Plain.* Nothing a condition has accumulated is lost when the life comes back to it, and nothing about how one now stands toward it is remembered by it. The history is in the place; the facing is in the one who arrives.

Theorem 9 (Locality)**.** *The selection depends only on the carried frame and the passages available at that facet. It does not and cannot read the holonomy, which is a property of circuits and is present* *at no vertex. Witnessed: two vessels with identical local passages but different holonomy dispatch identically on all carried frames.*

Theorem 10 (Dispatch is not idempotent over circuits)**.** *The selector is a fixed function; what changes is its argument. After a circuit the carried frame returns rotated by the holonomy, so the same position may dispatch differently on a later visit. Witnessed: over* 300 *carried axes the dispatch changes after a circuit in* 72% *of cases for a holonomy of* 1.40*, and in* 0% *for trivial holonomy.*

Corollary 3 (Invisible to a step, legible in a life)**.** *No single dispatch reveals the knot. But routes differ with the holonomy, and where the facets hold distinct invariants the interior history differs accordingly — witnessed, routes* $[0, 2, 1, 2, 0, 1, 1]$ *against* $[0, 0, 0, 0, 2, 2, 2]$ *and histories differing by* 0.25 *for holonomies* 0 *and* 3.04*. Where the facets are alike, routing leaves no trace at all.*

10 The bundle in motion

Proposition 13 (Transient degeneration, and what repairs it)**.** *The facets' interiors flow, and the passages glue them by isometries. A* generic *perturbed metric has no isometries at all:* $\mathbb{H}^3$ *has a six-dimensional isometry group and a generic nearby metric a zero-dimensional one. During a cathartic episode the gluing is therefore only approximate. What restores it is 3.5's own stability result — every perturbation decaying, the slowest at rate one — so the defect falls as* e^{-t}*:* $0.607, 0.368, 0.135, 0.018, 0.0003$ *at* $t = 0.5, 1, 2, 4, 8$*. The relational layer is exact between episodes and approximate during one.*

Corollary 4 (A dependency, and the knot's immunity)**.** *This layer is well posed only because the canonical form of 3.5 is an* attractor *and not merely a fixed point. The knot is nonetheless untouched throughout: a holonomy conjugacy class is discrete, and a defect that varies continuously and tends to zero cannot carry a representation from one class to another.*

11 Whether a frame persists at all

Theorem 11 (The corpus's own law aligns will and act)**.** *The core gives*

$$\dot{V} = \mu\,(aF + c\lambda) - \rho V, \qquad \dot{F} = \mu\,(aV + c\lambda) - \rho F, \qquad \mu = 1 - \tfrac{\sqrt{VF+\varepsilon}}{\Omega_P},$$

with the same *coefficient a coupling* $V \to F$ *and* $F \to V$*. The coupling matrix* $\left(\begin{smallmatrix} -\rho & \mu a \\ \mu a & -\rho \end{smallmatrix}\right)$ *is therefore* ***symmetric****, with eigenvectors* $(1, 1)$ *and* $(1, -1)$ *and eigenvalues* $-\rho \pm \mu a$*. Whenever* $\mu a > 0$ *the aligned mode is the less damped and dominates: witnessed,* $\cos(V, F) \to 1.000000$ *from every start tested, including* -0.8352*, under either equivariant placement of the scalar drive* $c\lambda$*.*

Corollary 5 (Orientation is transient; the knot is a phenomenon of division)**.** *Under the corpus's own law will and act converge in direction, so the frame degenerates and the knot goes latent.* ***The nonabelian layer therefore describes a transient condition, not a permanent structure.*** *Its phenomena are in force exactly while a life's willing and doing diverge, and lose their grip as the two come together. An antisymmetric coupling —* a *in one equation and* $-a$ *in the other — would give a persistent angle and a permanent frame; the corpus does not have that form.*

▷ *Plain.* Integration does not untie the knot — the passages are unchanged, and untying still waits on another — but it takes away what the knot acts on. As willing and doing come together, the turning has nothing left to turn.

Proposition 14 (The knot goes latent, not away). *Holonomy is a product of passage rotations and refers to no facet's V or F. It therefore remains defined when every frame has degenerated — but has nothing to act on. The knot persists in the passages while unrealised in the life, and becomes effective again as soon as some condition's will and act diverge.*

12 Passage and gate

Proposition 15 (The gate gives magnitude; the relation gives direction). *I_{gate} is a scalar influx. A scalar cannot specify a direction. Hence what is renewed at a passage is the strength of willing and acting, while the facing is inherited by transport. Were direction also given afresh, every passage would install a frame without memory, circuits would accumulate nothing, and no knot could exist — nor any continuity of a life with itself.*

Proposition 16 (Conditions for a knot to survive a passage). *The holonomy is preserved across a transition if degeneration is partial — the common axis surviving — if direction is inherited rather than created, and if the influx contributes magnitude only. Witnessed: a surviving axis returns from a three-cycle deviated by $78.99°$ against a holonomy of $80.36°$, unchanged by the impulse magnitudes along the way.*

13 The bridge

Theorem 12 (Interior labour cannot reach the knot). *Cleansing, the terminus reset, and folding act on the five retained invariants. The holonomy is registered in the three carried parameters. These are complementary halves of the same eight numbers, with no overlap; hence no operation on the first alters the second. An operation that changed the knot would have to add or cut a passage — a change of the graph, not of a state.*

▷ *Plain.* This is not a limit on the strength of interior labour, as though enough of it might one day suffice. Travelling and turning are different motions, and no distance corrects an angle.

IV. Status

14 What is established, and on what

Proved

The symmetry of the corpus's own will–act coupling and its consequence: alignment, frame degeneracy, latency of the knot. The 5+3 split and its inverse. The KAN assignment of state to A and connection to K, and the licensed structure $SO(3)\times\mathbb{R}$. The forced placement of the frame. Degeneracy under alignment. Subdivision invisibility. Non-additivity of rotation vectors. The invisible-knot theorem and the uniqueness of A_5 among finite perfect subgroups of $SO(3)$. The partial-invisibility ladder and the vacuity of nilpotent depth. The trace criterion. No false canonization. Locality of the dispatcher. The bridge theorem.

Witnessed numerically

The three Nielsen families (18, 10, 10). Two-fold trapping. The π_1 mechanism ($U(1)$ against $SU(2)$ on the same connection) and the separation of multiplicity from obstruction. Transient degeneration of the bundle and its e^{-t} repair. The alignment attractor and its rotational exception. The benign reconciliation landscape — unique minimum in every configuration tested, including torus lattices to 6×6 and three adversarial families, under two independent minimizers. Non-idempotent dispatch. Detectability from history.

Conditional, and on what

Everything nonabelian rests on two named commitments: that Voluntas and Factum are *directions* in 3.x, warranted by the corpus's own gloss and by its frozen-limit method but not stated in its text; and that their law is $SO(3)$-equivariant, which says only that willing and acting refer to no direction outside themselves. The reading of a facet as a full interior repurposes 3.5's "interior of a vessel".

Open

Answered (see the theorem above): the corpus's own law is symmetric, so orientation is transient and the knots of this volume are phenomena of a divided condition. What remains open is whether any vessel sustains an antisymmetric coupling, which would make orientation permanent. What makes activity move at all: no established mechanism requires it. Which facet receives it: answerable from the retained invariants, never from the knot. A proof of benignness. Whether invisible knots and half-turn passages arise in vessels, or are only permitted. The relation of this volume to the plural body of 4.0–4.6.

Not claimed

Every mathematical step here is standard — Hodge theory and gauge fixing, finite subgroups of $SO(3)$, holonomy and parallel transport, Nielsen equivalence. The contribution is transplantation and internal coherence, and no priority is claimed for any step.

Plain words

3.5 described what a soul is at a place: how much weight it carries there, how its willing and doing agree, how much wisdom it has won. All of that is *amount*, and amount is what a soul can measure about itself.

This volume is about the one thing it cannot. Willing points before it is measured; so does doing. Which way they both face is not among the measures — it is precisely what remains when facing is set aside. That facing is not stored in any part of a life; it travels with the living moment and is handed across each passage.

A passage is therefore two things at once, and cannot be split: where one goes and how one is turned in going. Choose the way and the turn is chosen. And turns do not simply add. So a life that comes round a circuit of its own conditions arrives where it began, with every measure exactly as it left them, and facing otherwise.

That turning is the knot. No single step shows it: the rule that chooses the next passage reads only what one carries and what is open, never the circuit. But where one is taken differs because of it, and over a life that difference is legible. And no labour within a condition reaches it: labour works on what is kept, and this is what is carried.

Interpretive reading

▷ *Modal (interpretive).* The measures a soul takes of itself are measures of amount, and orientation is not among them — it is what is left out when amount is taken. Yet the will is a direction before it is a quantity. So a life may be turned entire, and every reckoning it makes of itself read exactly the same. That is why the deepest wound cannot be found by examination however honest: examination reads what is kept, and this is not kept. It is why returning to a familiar condition, with everything about it unchanged, one may nonetheless face a way one did not before. And it is why such a knot waits for another: what is at issue is an angle, and an angle between two things belongs to neither. Only a life whose conditions are all alike would never be able to tell; and it would be spared nothing else.